

Assignment – 06: Replacement Metal Gate

This assignment explores the replacement gate method to define gate. The simpler gate-first method deposits the metal gate directly, followed by high-temperature processing. It is straightforward but limits the choice of metal due to thermal stability concerns. RMG, on the other hand, uses a temporary "dummy gate" to protect a high-k dielectric layer during high-temperature steps. This allows for a wider range of metal gates but adds complexities like extra processing steps and potential contamination issues.

Question 1: Replacement Gate

Replacement Metal Gate (RMG) utilizes a temporary "dummy gate" made of polysilicon. This allows for high-temperature processing steps without harming the actual metal gate, which is deposited later. The RMG process involves depositing a high-k dielectric layer first, followed by patterning and removal of the dummy gate. Finally, the desired metal is deposited and etched to define the final gate structure, enabling the use of a wider range of metals with superior electrical properties.

Figure 1 shows the Reference process for replacement metal gate.

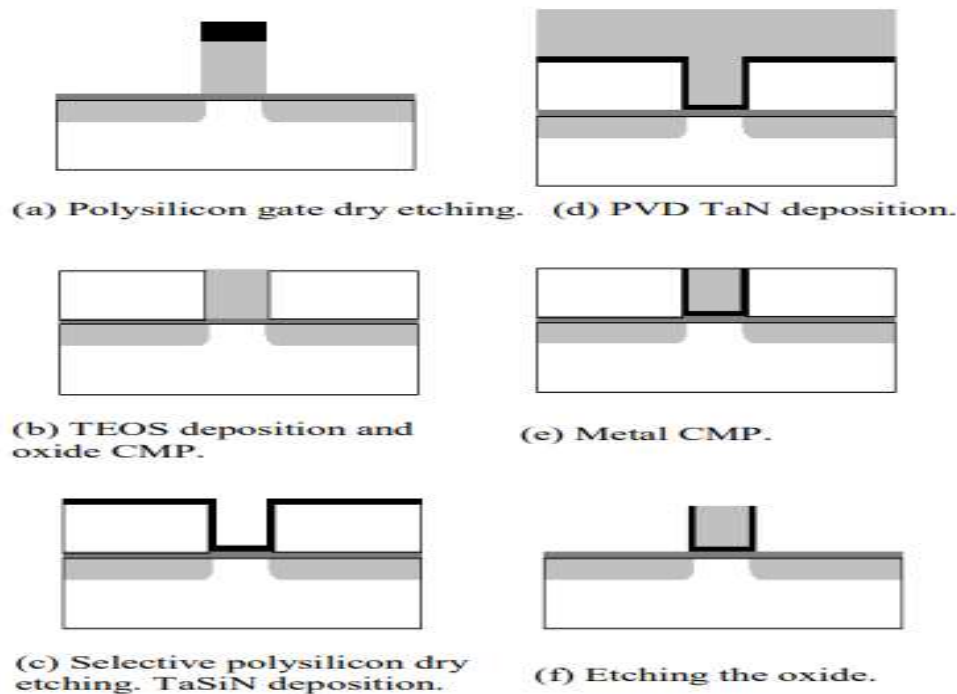


Figure 1 : Reference process for Replacement gate

Breakdown of the RMG process is as follows,

1. **Sacrificial Gate Formation:** A temporary gate structure, often made of a material like polysilicon, is formed, which will be there until we implant source and drain of our device.
2. **High-k Dielectric Deposition:** A layer of insulating material with a high dielectric constant (high-k) is deposited on the wafer.
3. **Gate Patterning:** The sacrificial gate is removed using methods like RIE, leaving behind a patterned high-k layer, which can be filled with gate metal.
4. **Metal Gate Deposition:** A thin metal layer is deposited first as wetting metal (TiN/TaN) and the gap is filled using some other metal (Al) to replace the sacrificial gate.
5. **Etching and Cleaning:** Excess metal is removed, and the wafer is cleaned.

The process flow followed for replacement gate process is shown by Figure 2.

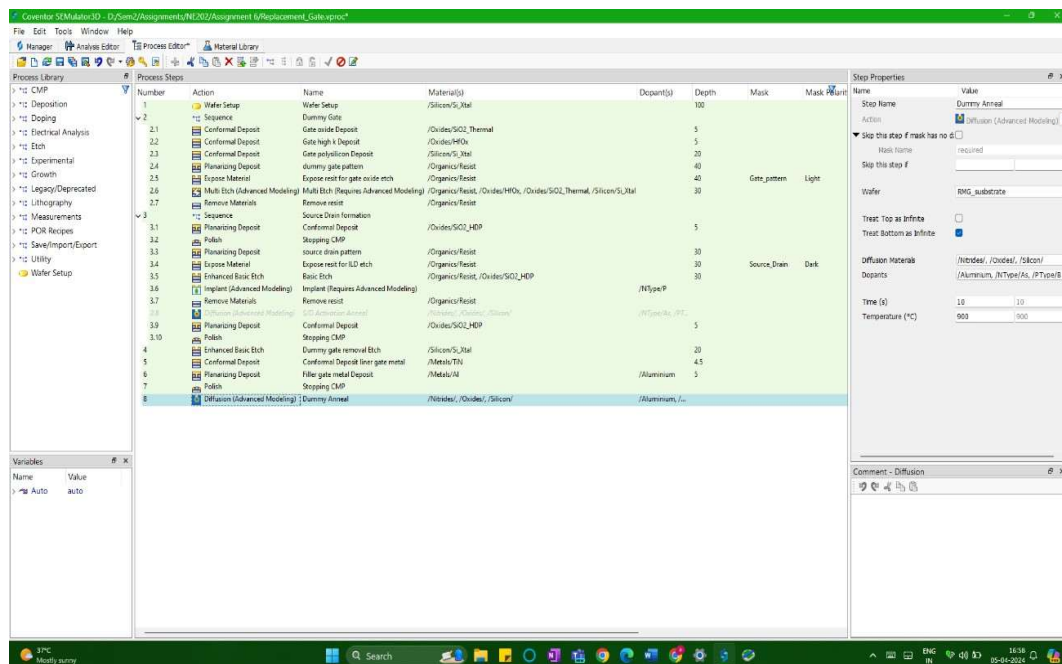


Figure 2 : Steps involved in simulation for replacement Gate process

Simulated results for replacement gate process is shown by Figure 3.

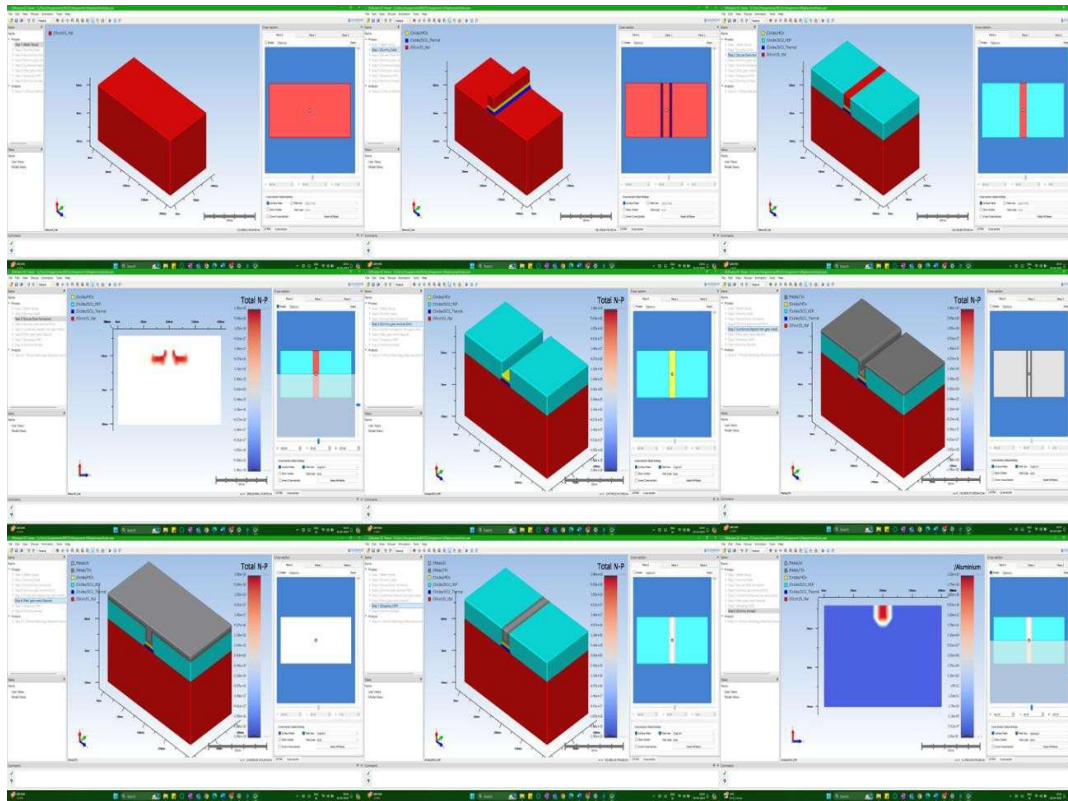


Figure 3 : Simulated Results for replacement gate process

We started with a P-type wafer, making polysilicon gate first after adding gate oxide and high dielectric material. An oxide layer is deposited and phosphorous is doped to make NMOS device. For gate, we etch away the temporary gate (as gate should be highly conductive) and deposits a liner of TiN as barrier for aluminium diffusion then the trench is filled with aluminium. And after polishing it the device is ready.

In the last step (Anneal) we can see aluminium diffusing up to a certain distance in device.

Question1(a)

This question asked us to find the average concentration of diffused aluminium in active region of device. Here active region is considered from (50,50,50) to (150,50,35).

The diffusion coefficient of Al in different material is given by Figure 4.

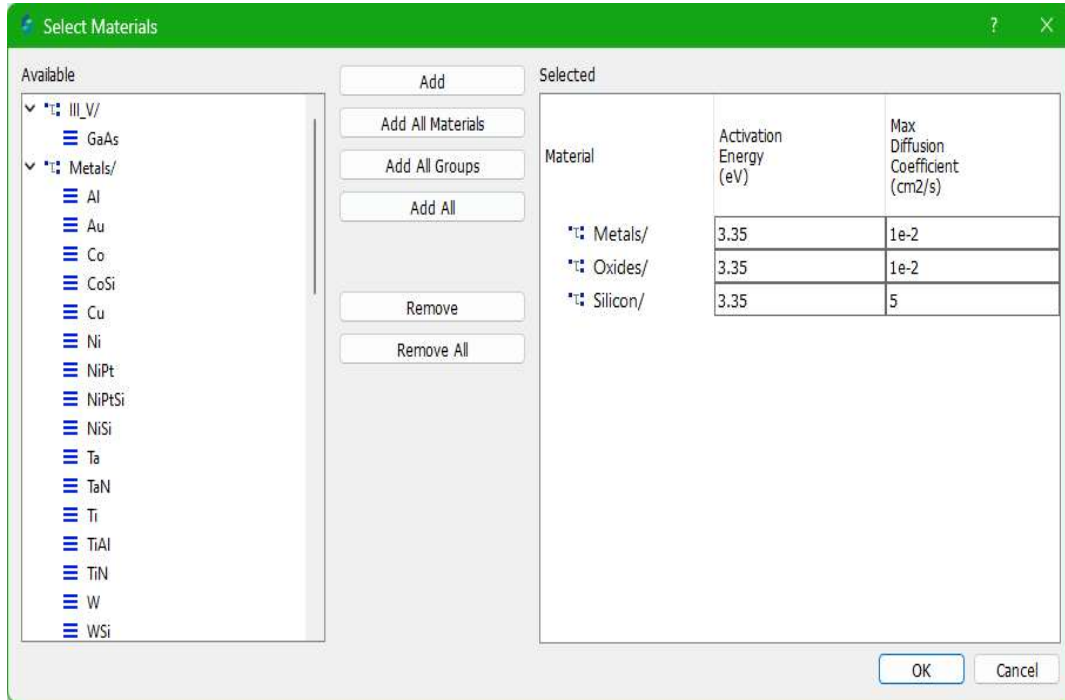


Figure 4 : Diffusion coefficients of Al

Diffusivity of Al is calculated as given in Table 1 for different material in SEMulator 3D.

Table 1 : Diffusivity of Aluminium

Sl.No.	Material	Diffusivity (cm ² /sec)
1.	TiN	4.04362e-17
2.	Silicon	2.02181e-14
3.	SiO2_HDP	4.04362e-17
4.	High-k dielectric (HfOx)	4.04362e-17
5.	SiO2_Thermal	4.04362e-17

The cross section of device after annealing (Figure 5) shows that aluminium is diffused in barrier layer and high k layer and oxide as well, but active region is untouched by it.

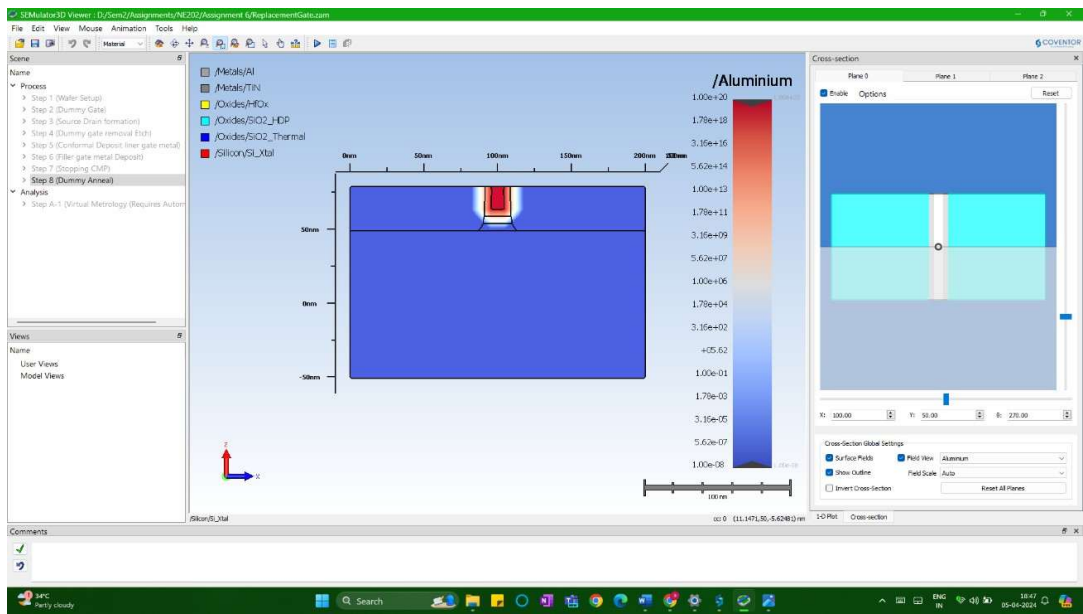


Figure 5 : Diffusion of Al in device

The average concentration of Al is 0 cm^{-3} as seen by Al profile below gate (Figure 6) in active region.

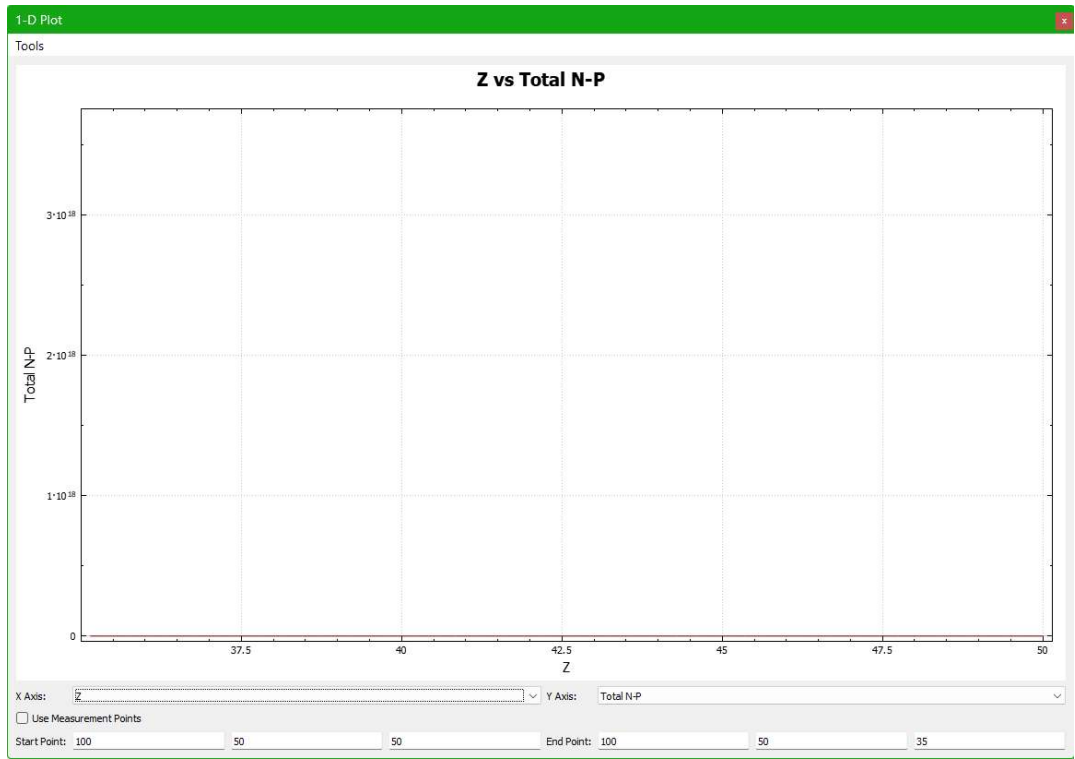


Figure 6 : Al Concentration vs Z

Question1(b)

In this part we must vary anneal temperature to find highest of temperature for annealing where metal contamination limit is reached in device active region.

- Metal contamination limit – $1 \text{ e}+12 \text{ cm}^{-3}$.
- Anneal time – 10 second.

From average concentration above the Al concentration is way below metal contamination limit. So increasing temperature for anneal time 10 sec as per Table 2 below,

Table 2 : Variation in Anneal temperature

Sl.No.	Anneal temperature (°C)	Al Concentration at interface (50nm) (cm^{-3})	Average Al Concentration (cm^{-3})
1.	900	0.00	0.00
2.	950	97.00 e-5	2.34 e-05
3.	1000	3.05 e+6	1.46 e+05
4.	1040	1.05 e+12	5.61 e+10
5.	1050	5.75 e+12	3.14 e+11

Data above shows increase in diffusion with increase in temperature as shown by Figure 7.

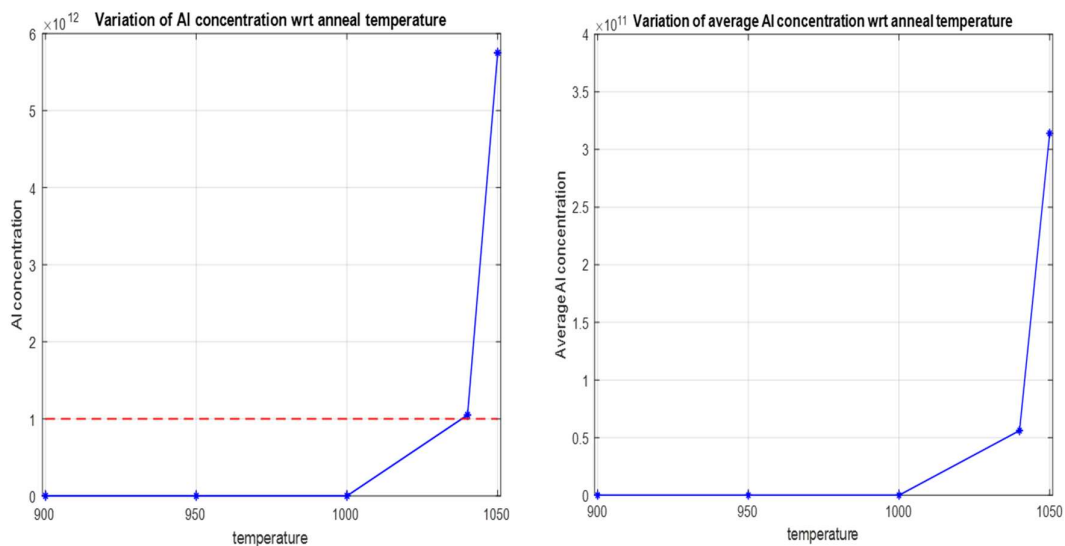


Figure 7 : Variation in Al concentration wrt to anneal temperature

Maximum temperature for which we can anneal before the contamination limit is reached is below **1040°C**, as interface Al concentration is **$1.05 \text{ e}+12 \text{ cm}^{-3}$** at interface and average concentration is less than that.

Simulated results for above data are shown by Figure 8.

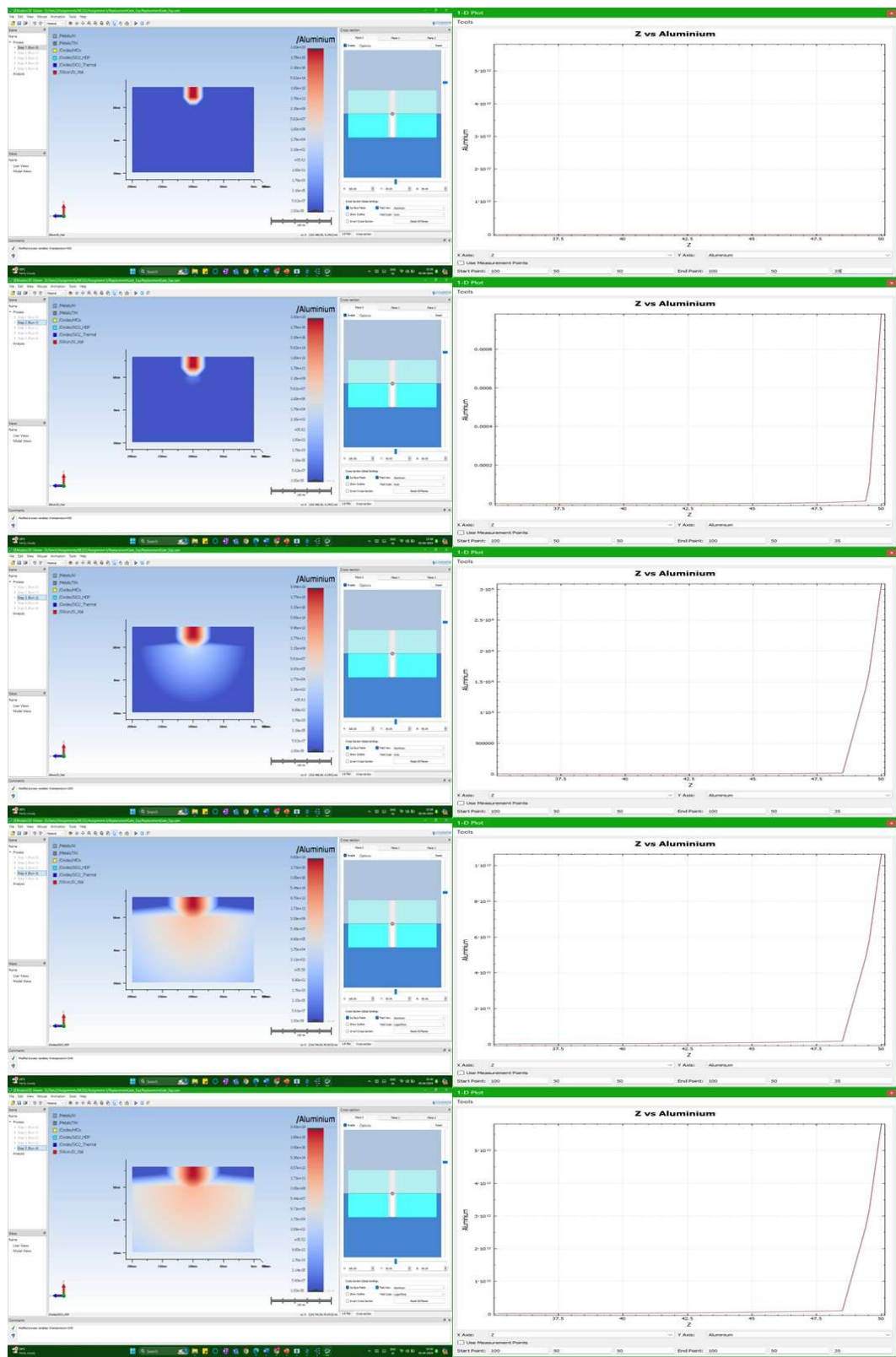


Figure 8 : Results for Table 2

Question1(c)

The added complexities for replacement gate process with respect to gate first process are given below –

- The major extra step added to the gate first process are **Step 4 to Step 7** in which polysilicon gate is etched away and Al gate along with wetting metal is deposited.
- Materials used for high dielectric layer in **Step 2.2** needs to be deposited before the high-temperature metal deposition in RMG. This requires careful selection of materials and process conditions to ensure compatibility.
- Here, we have not used the lift off process but usually to remove wetting material we use lift off. It might seem simpler as we only have to dissolve the photoresist layer which will take unwanted metal away, but this process might leave behind some residual photoresist material, which can contaminate the final device. In the current process we are polishing (CMP) surface after metal deposition **Step 7**.

Question 2: Metrology of Replacement Gate

Question2(a)

The process parameters that affect the replacement gate process are as follows-

- **Multi etch step (Step 2.6)** – High lateral etch ratio in multi etch step ensures precise control over the gate dimensions. This allows for the creation of very thin gates, crucial for scaling down transistor sizes. While low lateral etch ratio can lead to excessive undercutting of the gate material.
- **CMP (Step 3.10)** – The wafer surface might have some roughness due to the inherent nature of thin-film deposition processes. CMP is used to smoothen these surfaces and achieve a fine finish within a certain roughness tolerance ($< 1\text{nm}$). To ensure that surface is uniform and smooth for upcoming steps like lithography, we need to perform CMP in a controlled manner. If surface is rough photoresist adhesion and patterning fidelity can be affected negatively.
- **Etching of temporary gate (Step 4)** –The removal process for the dummy gate needs to be selective towards the gate material and avoid damaging the underlying dielectric. Also it should be highly uniform in removing polysilicon, otherwise either it will leave polysilicon behind or etch the layer beneath it.
- **Wetting metal (TiN) thickness (Step 5)** – The thickness of the TiN liner impacts etch barrier strength, gate profile, and gate resistance and the final shape of the etched gate.
- **Metal deposition (Step 5 and Step 6)** – The metal deposition process parameters like temperature and pressure affect the film quality, morphology, and adhesion to underlying layers. The deposition should be conformal for high aspect ratio trenches to prevent void formations.

Question2(b)

This part of assignment asks to see the variation of gate critical dimension with respect to etch parameters.

In the process flow there are having etching at three places (Figure 9), first to make dummy gate, second to implant source and drain and lastly to remove dummy gate. Out of this three the most critical is the first one when we deposit polysilicon to make temporary gate because it will define gate dimensions. So here we are varying step 2.6 multi etch.

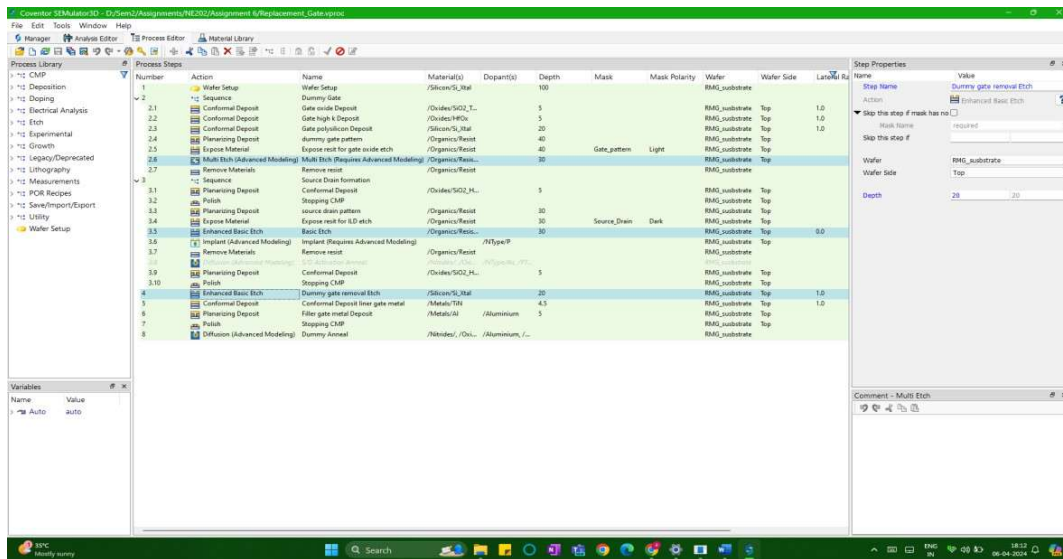


Figure 9 : Etch steps in process flow

For using virtual metrology we defined mask as shown in Figure 10.

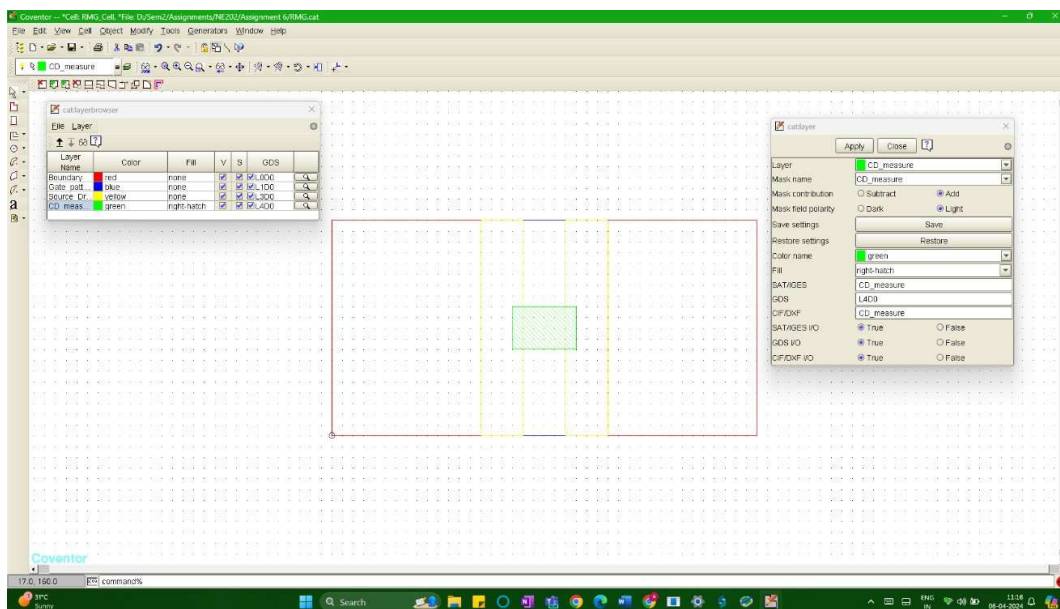


Figure 10 : Mask for measuring gate CD

After creating mask, we defined parameters for critical dimension measurement as shown by Figure 11.

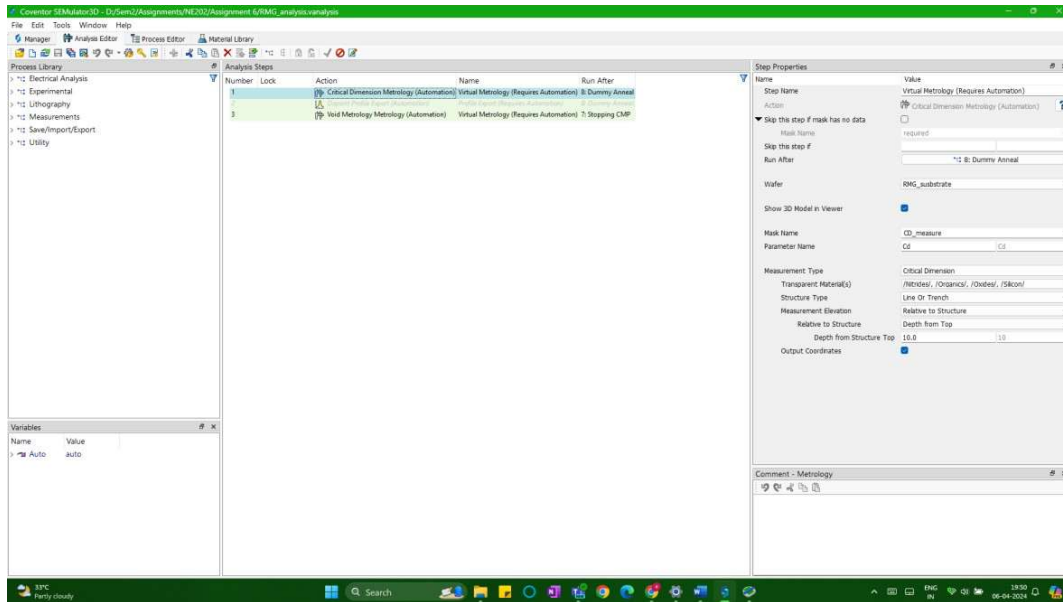


Figure 11 : Parameters for CD measurement

The lateral etch ratio is varied from 0.025 to 0.2 for every material as given in Table 3.

Table 3 : Critical dimension Variation

Sl.No.	Lateral Etch Ratio	Gate Critical Dimension
1.	0.025	18.718
2.	0.050	17.476
3.	0.075	16.234
4.	0.100	14.961
5.	0.125	13.656
6.	0.150	12.367
7.	0.175	11.054
8.	0.180	10.789
9.	0.186	10.468
10.	0.190	10.257
11.	0.195	10.023
12.	0.200	09.766

The data extracted from SEMulator 3D expeditor is plotted in Figure 12 which shows linear decrease in gate critical dimension with increase in lateral etch ratio which can be predicted as the lateral etch ratio is more the more material will be etched away from sides then top.

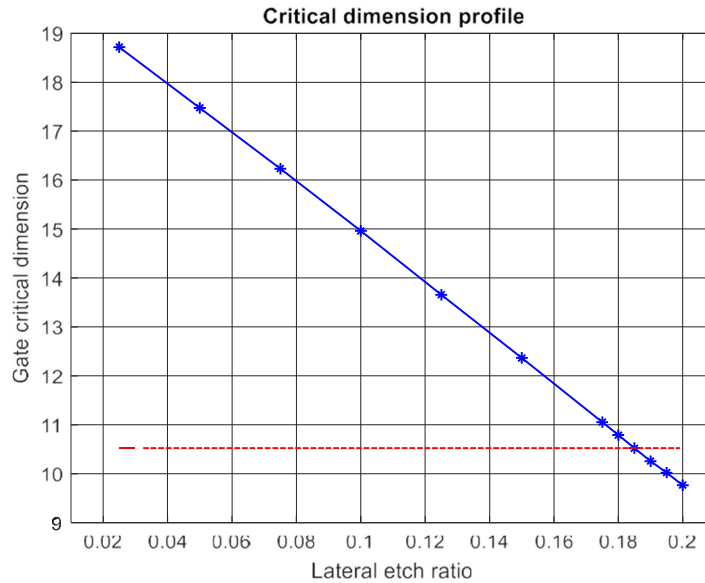


Figure 12 : Critical dimension Profile

The simulated results shown in Figure 13. The void formation can be seen for lateral etch ratio from **0.186 to 2.0** i.e. last four cases.

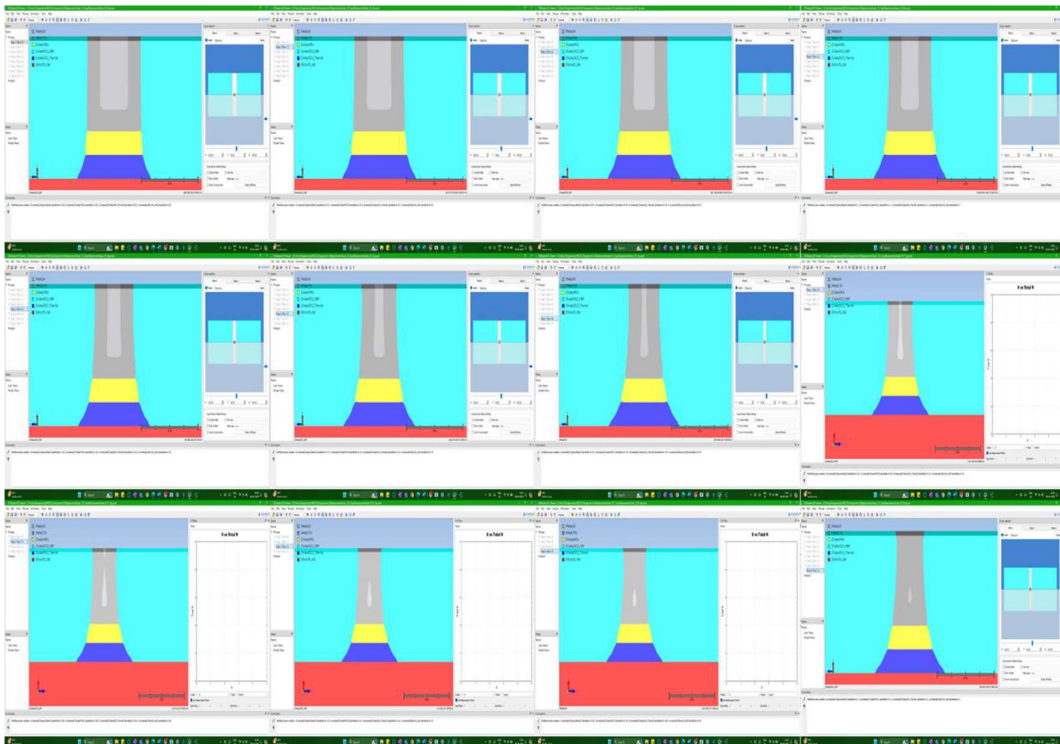


Figure 13 : Simulated results for CD variation

Question2(c)

The process fails when a void is formed this happens when the TiN layer deposited on corners touches each other or in other words top of gate length is less than bottom gate length. It becomes a concern when we fabricate devices with high aspect ratio. It can lead to variation in device performance and unreliable operations.

In this process flow the starting of void formation is seen when the **lateral etch ratio is 0.186 and above**.

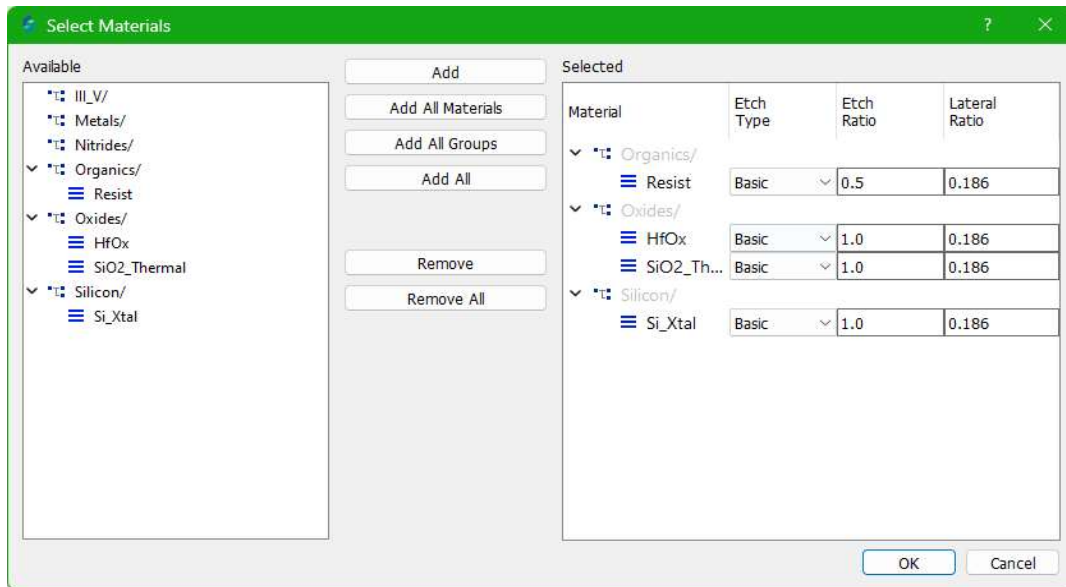


Figure 14 : Lateral etch parameters

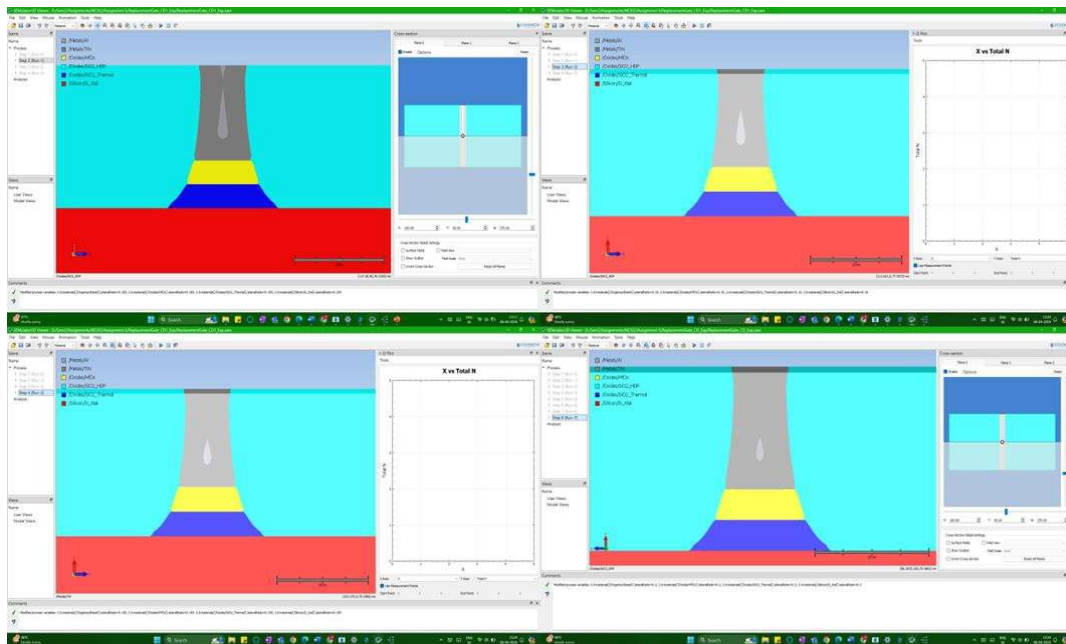


Figure 15 : Void formation due to lateral etch ratio

Another step where a void can form is when we deposit metal non conformally. So here we changed conformal deposition of TiN to visibility. Keeping lateral ratio constant at 5 and source sigma constant at infinity while varying isotropic ratio from 0.2 to 0.3. we can see void forming when **isotropic ratio is 0.225**.

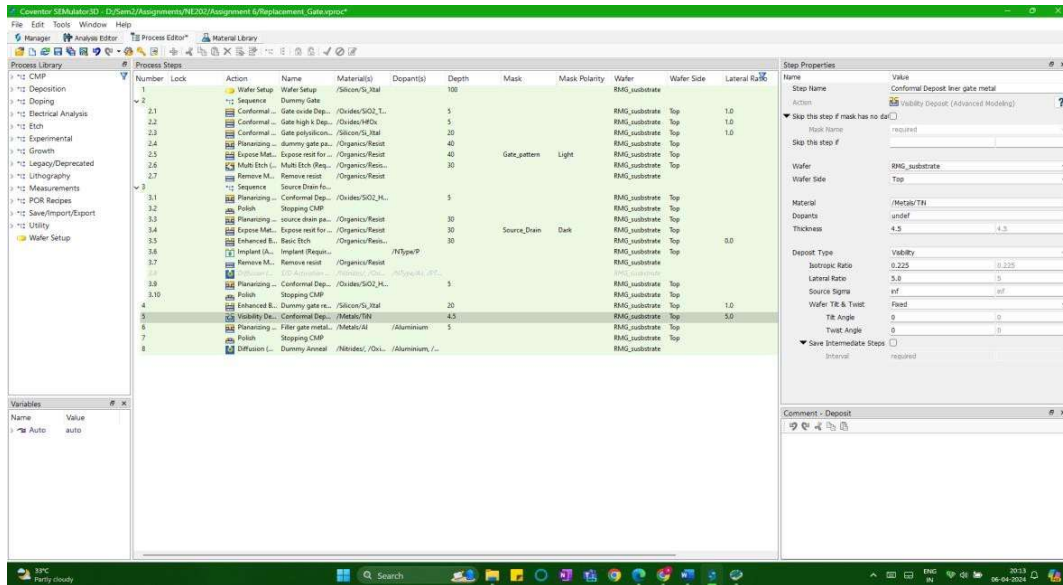


Figure 16 : visibility deposition of TiN

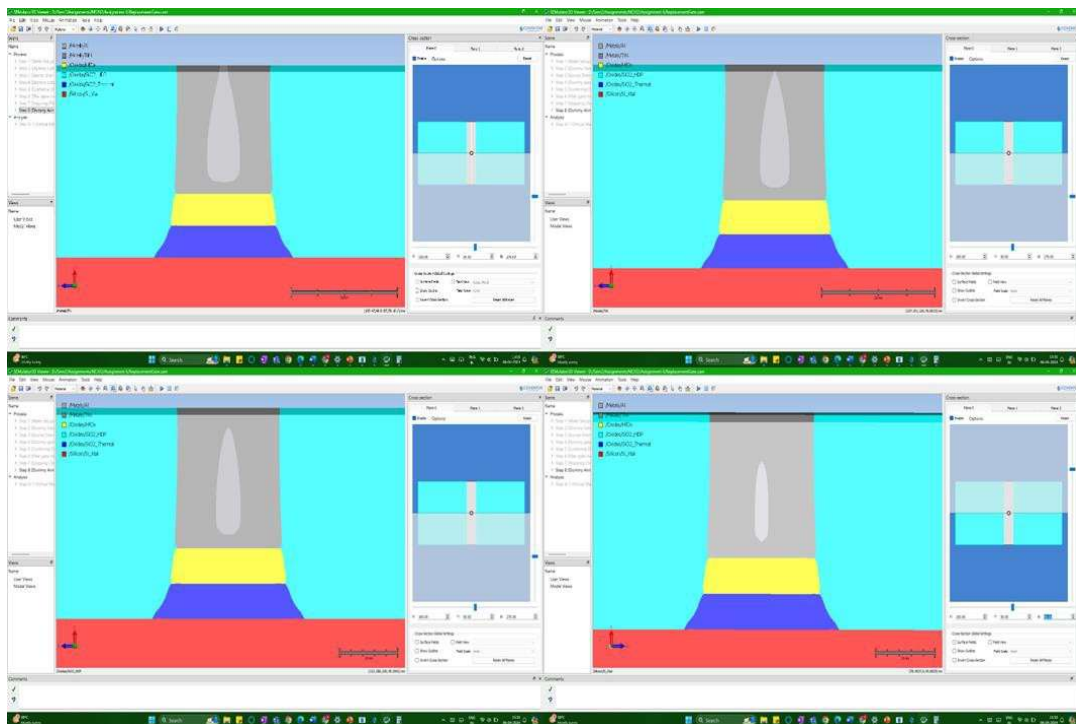


Figure 17 : Void formation due to metal deposition technique

Question2(d)

This part is about calculating the void dimensions formed during processing using void metrology in virtual metrology.

The parameters for void metrology is shown in Figure 18.

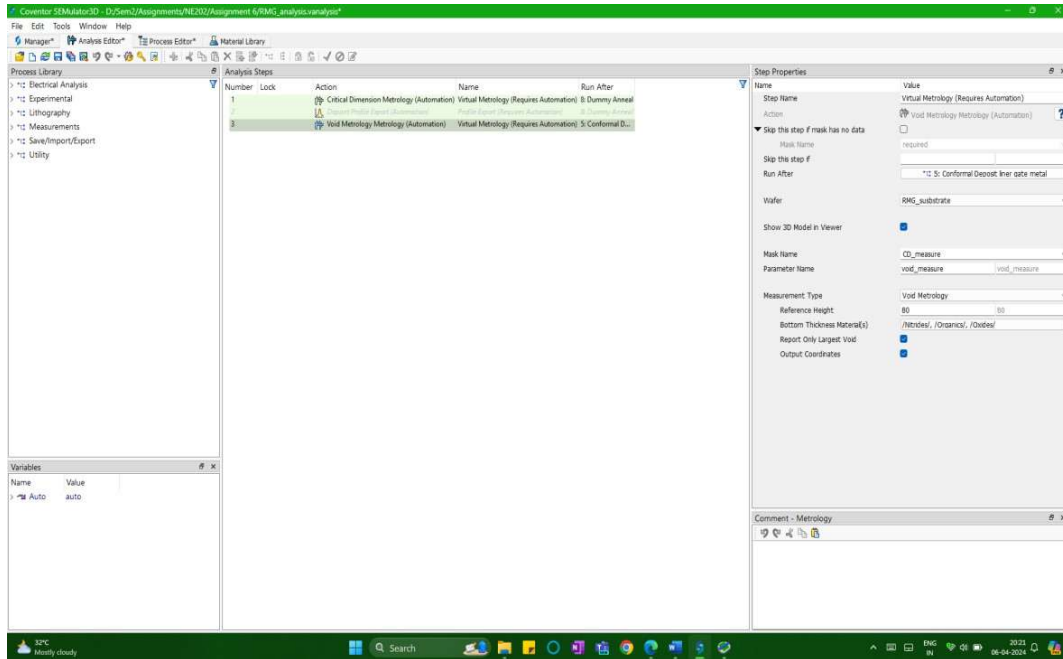


Figure 18 : Parameters for void metrology

The dimension for void is shown in Figure 19.

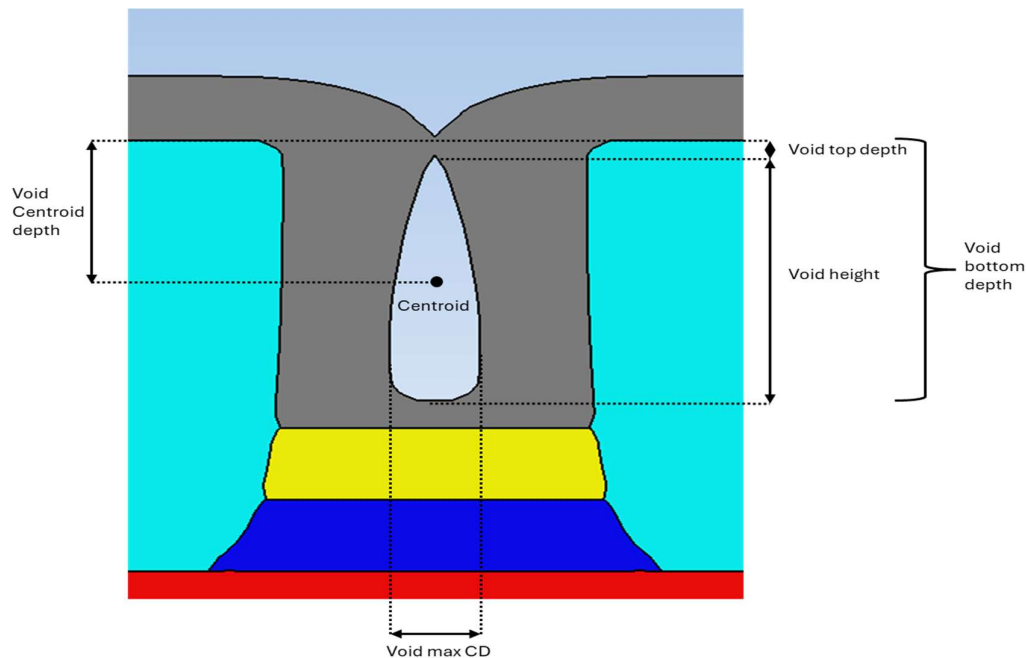


Figure 19 : Void dimension

The void dimensions measured in metrology is given by Table 4 and simulated results are shown in Figure 20 and Figure 21.

Table 4 : Void metrology results

Parameter	Void due to lateral etch ratio (nm)	void due to metal deposition (nm)
Void max CD	2.212	5.172
Void depth	5.285	1.000
Void height	10.214	17.118
Void area	12.514	65.031

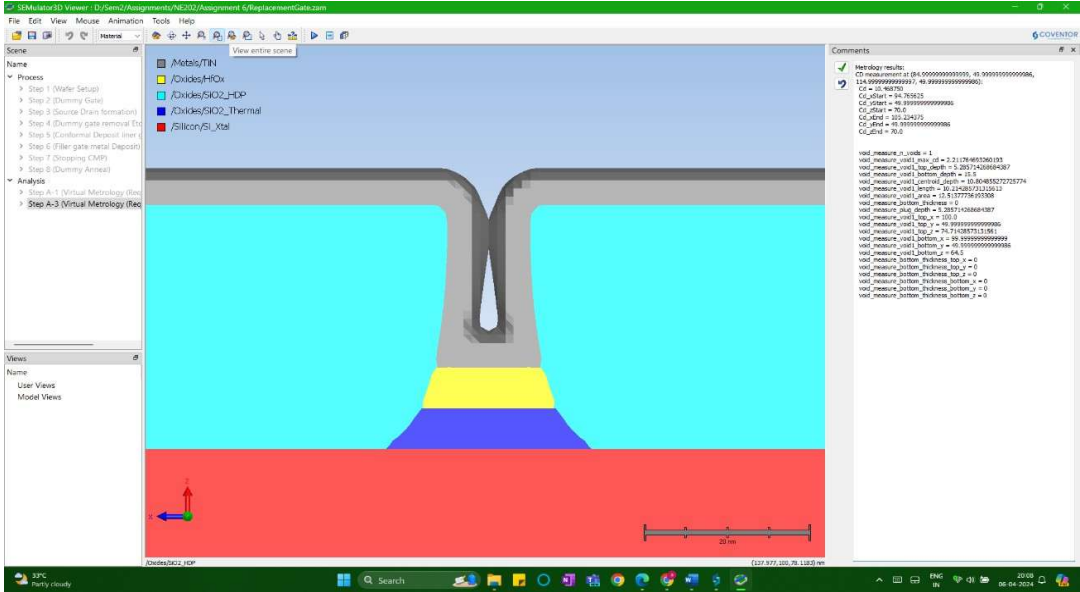


Figure 20 : Metrology results for void due to lateral etch ratio

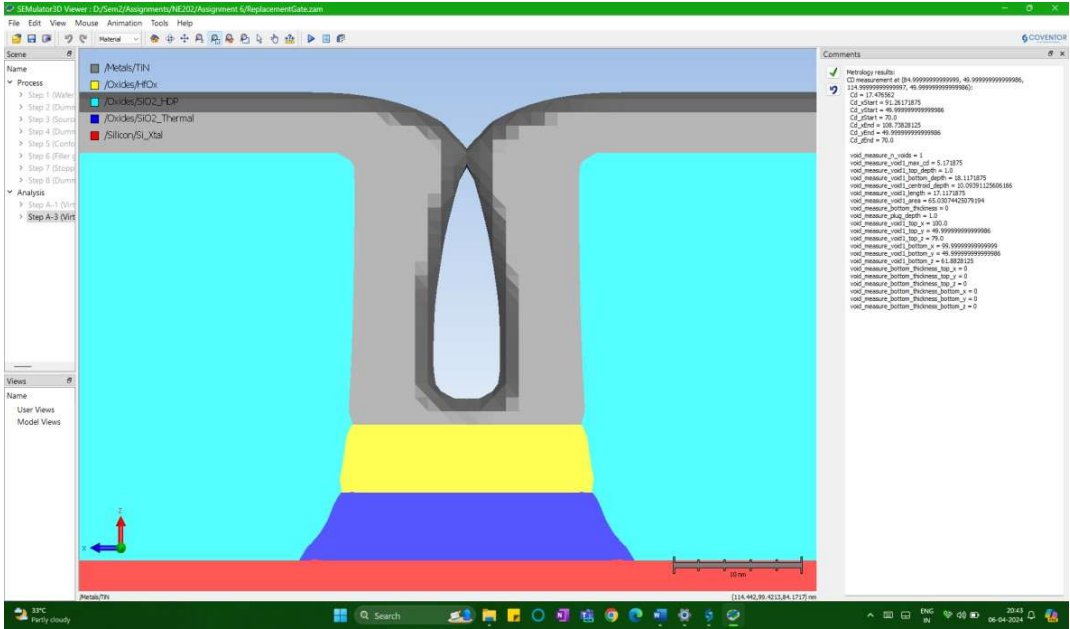


Figure 21 : Metrology results for void due to metal deposition

Conclusion

Analysing the Replacement Gate process revealed a trade-off between diffusion control and complexity. While RMG offers flexibility in gate metals due to the high-temperature source/drain anneal, it raises metal diffusion concerns. Simulations helped identify the maximum safe anneal temperature to avoid excessive diffusion. Compared to the simpler gate-first approach, RMG adds complexity with sacrificial polysilicon gate steps and potentially different etch chemistries. Understanding critical parameters like etch ratios, liner properties, and deposition conditions is crucial for optimizing RMG with the help of virtual metrology we can predict gate critical dimension, identifying void formation, and quantify their impact. This knowledge allows engineers to balance device characteristics with diffusion control and process complexity.

References

- [1] Metal Gate NMOSFETs With TaSiN / TaN Stacked Electrode Fabricated By A Replacement (Damascene) Technique James Pan, Minh-Van Ngo, Christy Woo, Jung-Suk Goo, Paul Besser, Bin Yu, Qi Xiang, Ming-Ren Lin Technology Research Group, Advanced Micro Devices, Inc., Sunnyvale, CA 94088, USA.
- [2] Diffusion of Metals in Silicon Dioxide J. D. McBrayer, R. M. Swanson,* and T. W. Sigmon* Stanford Electronics Laboratories, Stanford, California 94305
- [3] Help from SEMulator 3D.